CE462
Project 2

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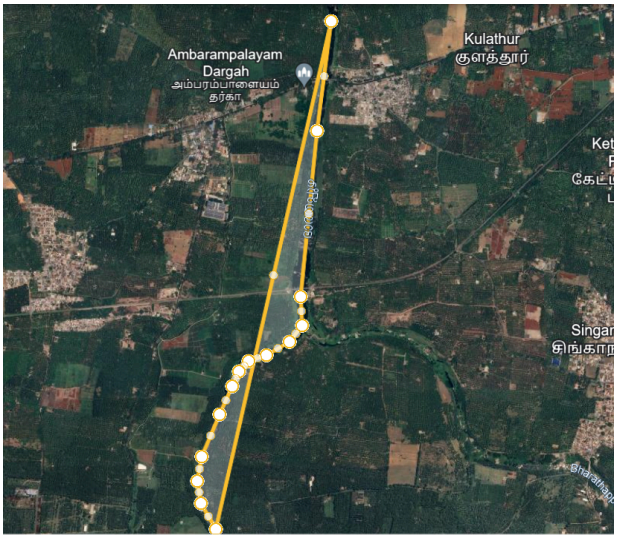
Question 1



Parameters for Rosgen Stream Classification			
Average Depth	218.579506	m	(level)
Bankful Surfac	40	m	
W/D	0.1829997731	<12	
Channel Lengt	3419.22	m	
Valley Length	3199.81	m	
Sinuosity Ratio	1.068569696	<1.1	Straight
Mean Sediment	0.12	mm	
Rosgen Stream	A		

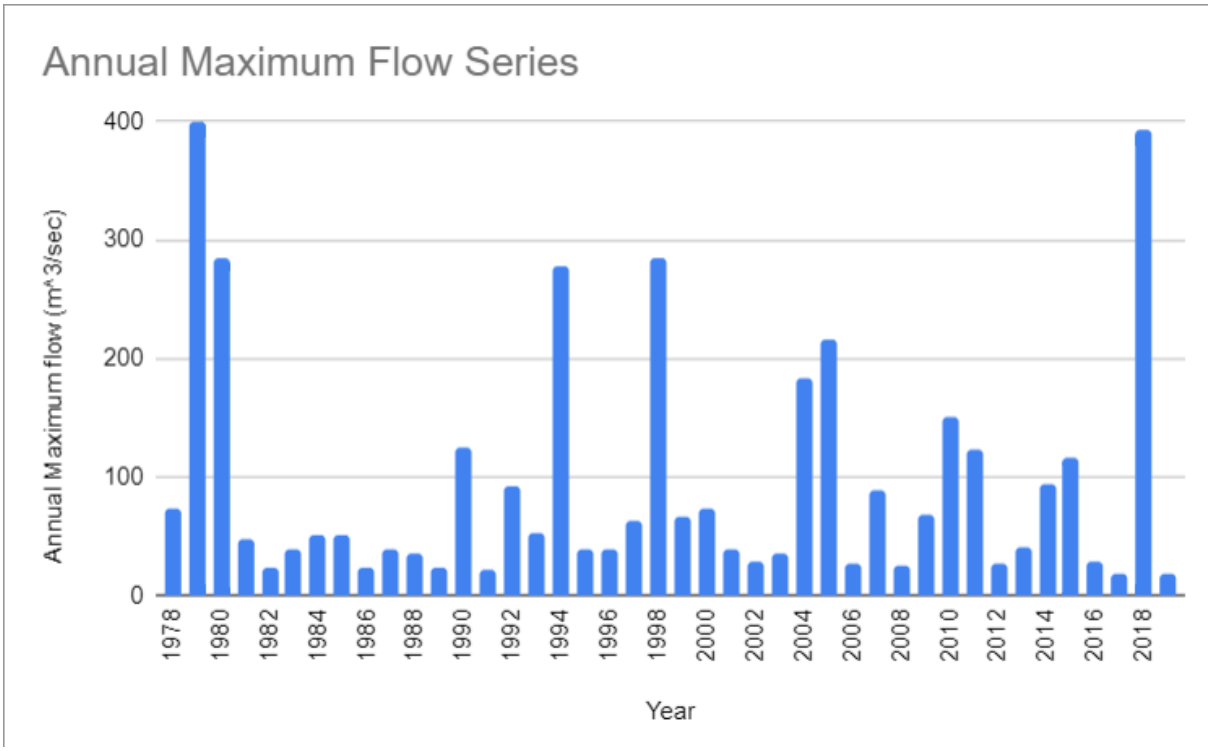
Part a:

Commenting on the site:

- ➔ The stream is straight at the location.
- ➔ The channel is narrow and banks are firm.
- ➔ The approach road is free from obstacles.
- ➔ There is a proximity to a direct alignment of the road to be connected.
- ➔ There are no sharp curves in the approaches.
- ➔ Rosgen stream is classified to be straight and under class A.

Therefore, the site is perfectly suitable and safe for bridge construction.

Part b:
The plot of Annual Maximum Flow series:



After the parameter estimation of the distributions log-normal, exponential, gamma and gumbel, to test the goodness of fit, chi-square test is performed.
Log- Normal has failed the test.
In the model selection criteria using AIC and BIC, Exponential had the minimum values and the best fit.
A total number of **220 missing records of flow values and 108 missing records of level values** are found in the entire data.
The data is downloaded from all the agencies and the unavailable data is also missing in CWC (Central Water Commission). Hence, retrievement became impossible here.
So, the zero flow data and the blank values are interpolated such that base flow is preserved and the average value of the non zero flows is preserved.

There are outliers in the data and they are eliminated and not considered as annual maximum flow

Generally bridges are constructed with return period 50 years, and hence the design discharge was calculated as 300 m^3/sec

Part c, Part d and Part e are below in the notes

Part (c):

Effective Waterway Length of a Bridge : W

Using Lacey's Regime theory,

$W = C\sqrt{Q}$, Discharge $Q = 300\ m^3/sec$

$W = 4.8\sqrt{300}$

$\boxed{W = 82.92m}$

Part (d)

Mean sediment diameter $d_m = 0.17$ mm

(K_{sf}) Silt factor $= 1.76\sqrt{d_m}$

$= 1.76\sqrt{0.17}$

$= 0.725$

Catchment area of Amaravati Basin = 8380 km^2

lies between 5000 - 25000 km^2

Q is increased by 20%

$Q' = 1.2 \times Q = 1.2 \times 300 = 358.16\ m^3/s$

α = Lacey's P

Normal scour depth, $D = 0.473\left(\frac{Q'}{K_{sf}}\right)^{1/3}$

$= 0.473\left(\frac{358.16}{0.725}\right)^{1/3} = 3.74\ m$

Maximum scour depth (IRC:78)

for piers, $D_{max} = 2D = 7.48m$

for abutments, $D_{max} = 1.27D = 4.75m$

Part (e) Discharge = 300 m^3/sec between 30-300 cumecs

IRC5 C106.2 ⇒ Minimum Vertical Clearance = 900mm

IRC5 C107 ⇒ freeboard should be more than 1.75m

∴ freeboard = 1.80m

Question 2:

Part a:

Maximum monthly demand came out to be 27645769.37 m^3

Using the sequent peak algorithm for all the 42 years daily discharge data, Live Storage Capacity is 985903089.1 m^3.

If the reservoir is to be designed for 100 years, the required dead storage capacity for the given sediment load characteristics is 3467891.744 m^3. (Using C/I method)

Question 3

Question - 3

Spillway and Stilling Basin Design

Part (a)

We have to calculate discharge for 100 year return period

$T = 100$ years, $p = \frac{1}{T} = 0.01$

$F(z) = 1 - p = 0.99$

Interpolating $F(z)$ & z $\quad 0.9890 \rightarrow 2.32$

$0.9901 \rightarrow 2.33$

$z = 2.326$ Standard Normal

$\therefore z = \frac{x - \mu_x}{\sigma_x}$ As the best fit is exponential & the data follows this distribution

$x = \sigma_x \cdot z + \mu_x$

$= 99.5 \times 2.326 + 94 = 325.54 \ m^3/sec$ (Design Discharge)

Part (b)

Maximum allowable water depth = 54.8 m

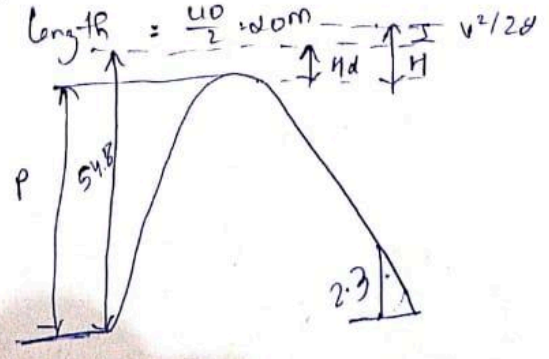
Slope (n) of the straight channel = 2.3

Tailwater level = 6.7 m

Existing river width (calculated from Google earth) = 40 m
- perpendicular to the river flow.

Design discharge = 325.54 m^3/sec

Effective crest length $= \frac{40}{2} = 20 m$



$Q = 325.5\ m^3/s \quad L_e = 20m$

Assuming $P/H_d > 3$, $C = f(P/H_d) = 0.74$

$\therefore \quad Q = \frac{2}{3}\sqrt{2g}\, C L_e H^{3/2}$

$325 = \frac{2}{3}(2 \times 9.81)^{0.5} \times 0.74 \times 20 \times H^{3/2}$

$H = 3.8\ m$

$v = \frac{Q}{L_e \times 54.8} = \frac{325.5}{20 \times 54.8} = 0.29$

$\frac{v^2}{2g} = \frac{(0.29)^2}{2 \times 9.81} = 0.0044$

$H = H_d + \frac{v^2}{2g} \Rightarrow 3.8 = H_d + 0.0044$

$H_d = 3.7956$

$P = 54.8 - H_d = 54.8 - 3.7956$

$\boxed{P = 51.004}$

Checking the assumption $\frac{P}{H_d} = 13.43 \geq 3$

Our Assumption is Valid.

Now, Profile estimation: Using Table (IS 6934:2014)

when $\frac{P}{H_d} = 13.43 \Rightarrow \frac{A_1}{H_d} = 0.28 \quad \frac{B_1}{H_d} = 0.165 \quad k_2 = 2$

$\frac{A_1}{3.7956} = 0.28 \Rightarrow A_1 = 1.107$

$\frac{B_1}{3.7956} = 0.165 \Rightarrow B_1 = 0.65$

$\therefore$ Upstream quadrant profile

$\frac{x_1^2}{A_1^2} + \frac{y_1^2}{B_1^2} = 1 \Rightarrow \frac{x_1^2}{(1.1)^2} + \frac{y_1^2}{(0.65)^2} = 1$

∴ Downstream quadrant profile:

$$X_2^{1.85} = k_2 H_d^{0.85} Y_2 \quad , \quad k_2 = 2$$

$$X_2^{1.85} = 2 H_d^{0.85} Y_2$$

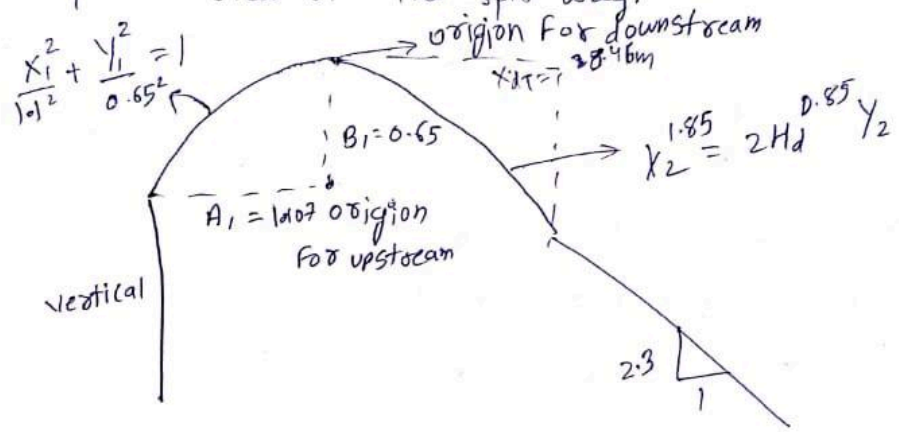
Also, $X_{DT} = 0.485 H_d (k_2 \alpha)^{1/0.85}$

where $k_2 = 2$, $\alpha = 2.3$ $\quad \because n = 2.53$

$$X_{DT} = 0.485 \times \cancel{8.1165}\ 3.7956 \times (2 \times 2.3)$$

$$= 8.46 \text{ m}$$

∴ The profile view of the spill way:



Part (c)

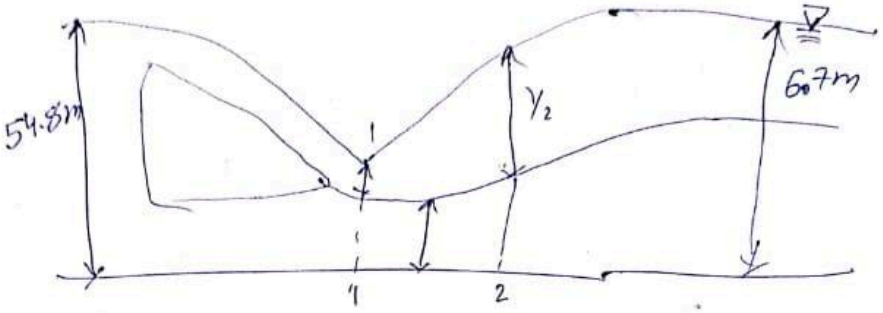
Discharge = 325.545 m^3/sec

Width of spillway = Effective crest length = 20m = b

Water surface elevation on tailwater = ~~6.9~~ 6.7 m

in the reservoir = 54.8

Assuming 10% loss of hydraulic head in the spillway channel



Energy equation between 0 and 1

$$54.8 = E_1 + X + 0.1(54.8 - 6.7)$$

$$E_1 = 54.8 - X - 4.81$$

$$E_1 = 49.99 - X$$

$$Y_1 + \frac{Q^2}{2gb^2Y_1^2} = 52.24 - X \longrightarrow (1)$$

Momentum equation:

$$\frac{Y_2}{Y_1} = \frac{1}{2}\left[\sqrt{1 + 8Fr^2} - 1\right]$$

$$\frac{4.9 - X}{Y_1} = \frac{1}{2}\left[\sqrt{1 + 8\frac{Q^2}{b^2 g Y_1^3}} - 1\right] \longrightarrow (2)$$

$$Y_1 + \frac{325.5^2}{2 \times 9.81 \times 20^2 \times Y_1^2} = 52.24 - X$$

$$Y_1 + \frac{13.5}{Y_1^2} = 52.24 - X \longrightarrow (3)$$

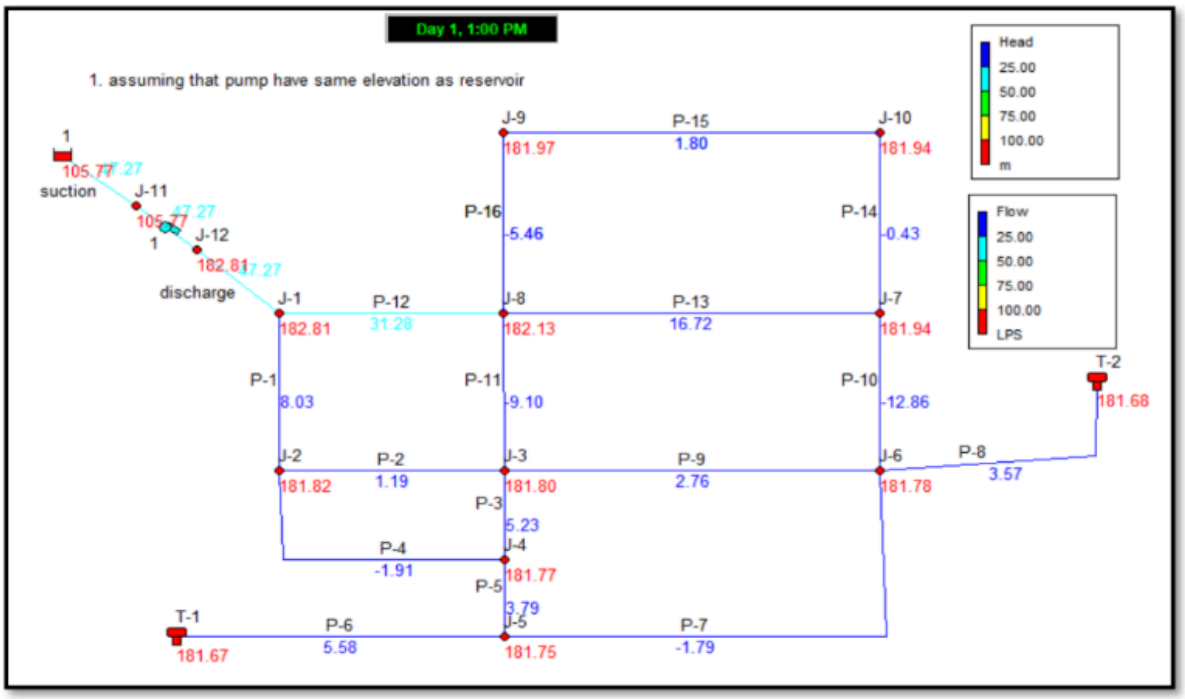
$$\frac{4.9 - X}{Y_1} = \frac{1}{2}\left[\sqrt{1 + 8 \times \frac{9325.5^2}{20^2 \times 9.81 \times Y_1^2}} - 1\right]$$

$$\frac{4.9 - X}{Y_1} = \frac{1}{2}\left[\sqrt{1 + \frac{2399.38}{Y_1^3}} - 1\right] \longrightarrow (4)$$

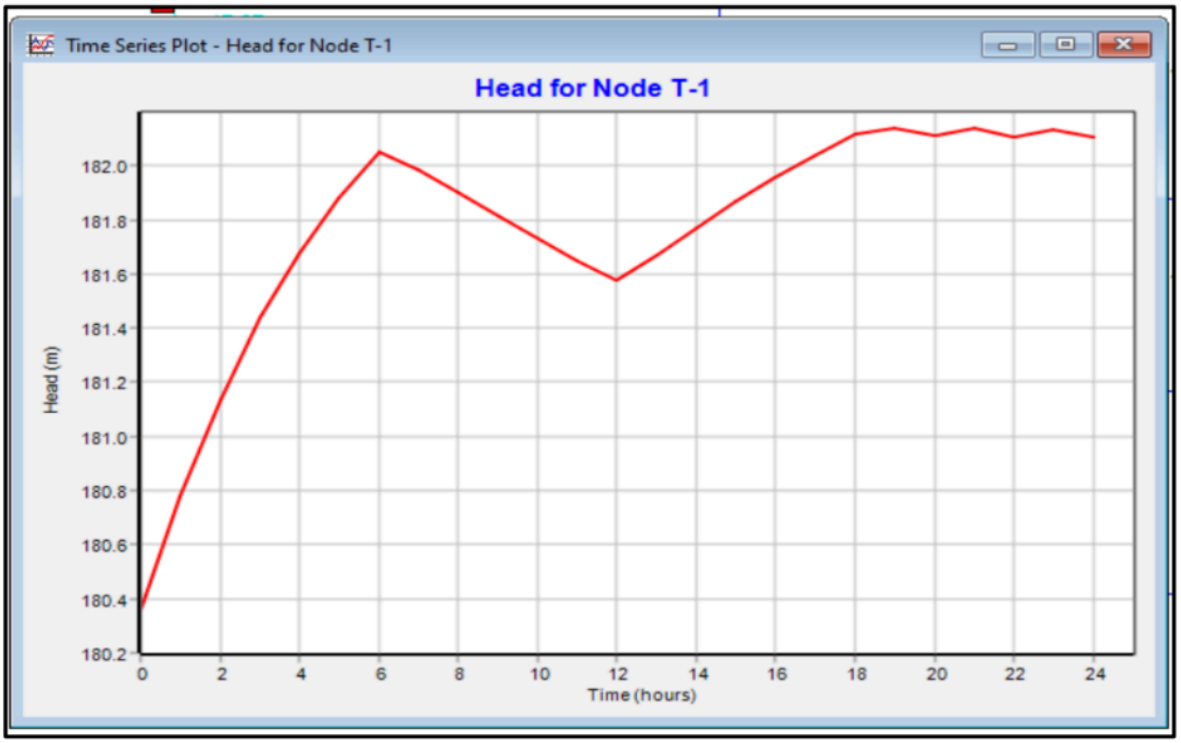
The equation (4) is non. linear & can be hard to solve equation (3)

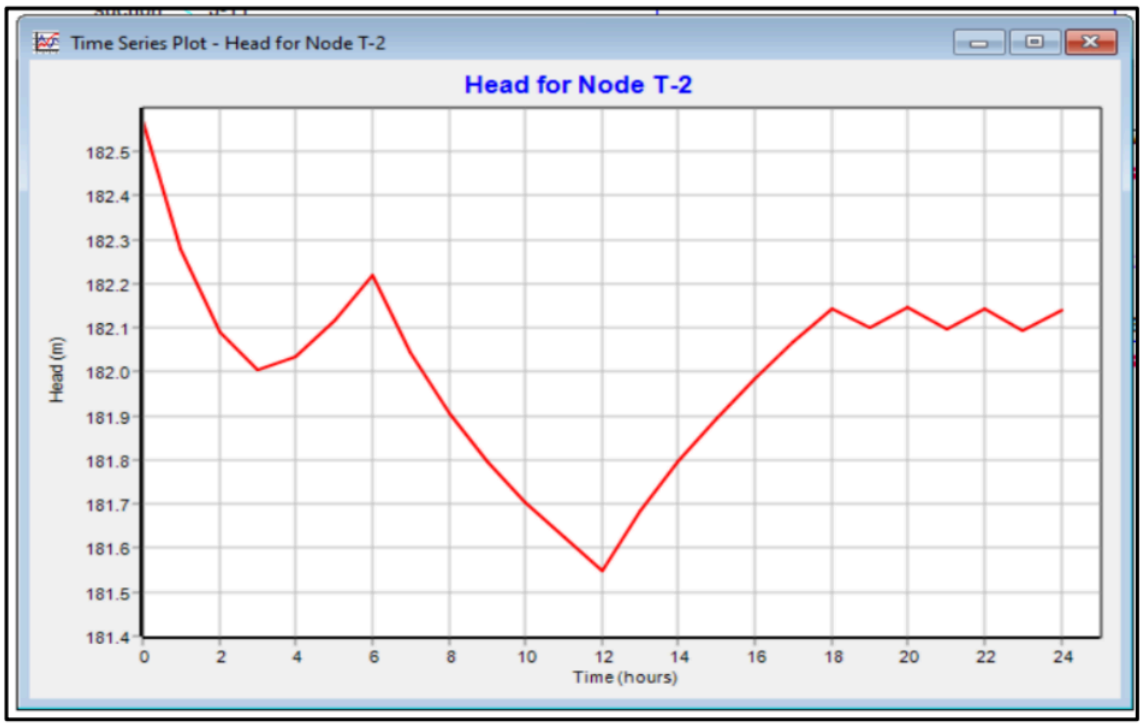
Question 4

The network is made on epanet software and plots are taken

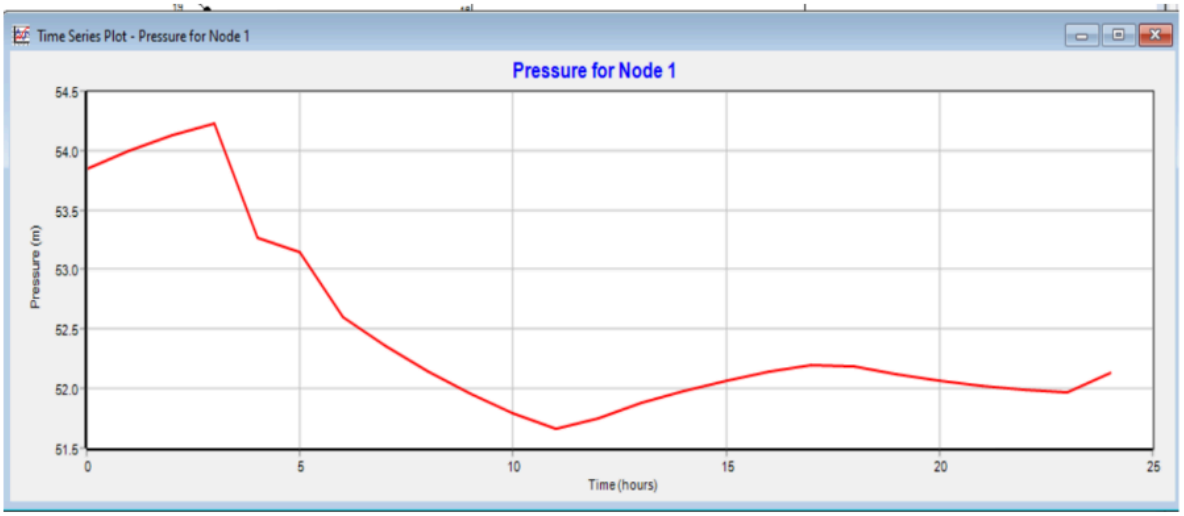


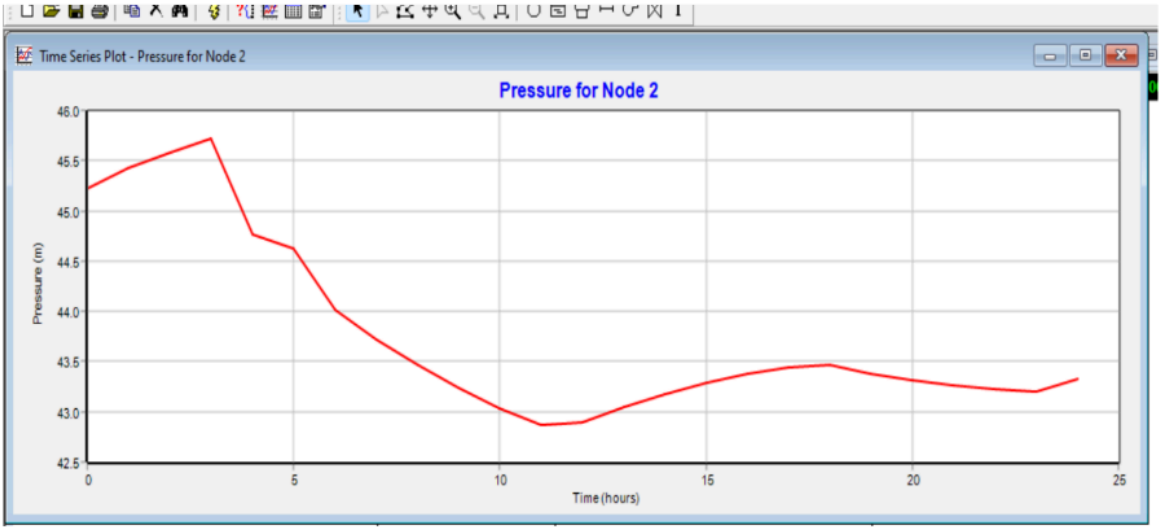
(a) time-series plot of HGL for the tanks.

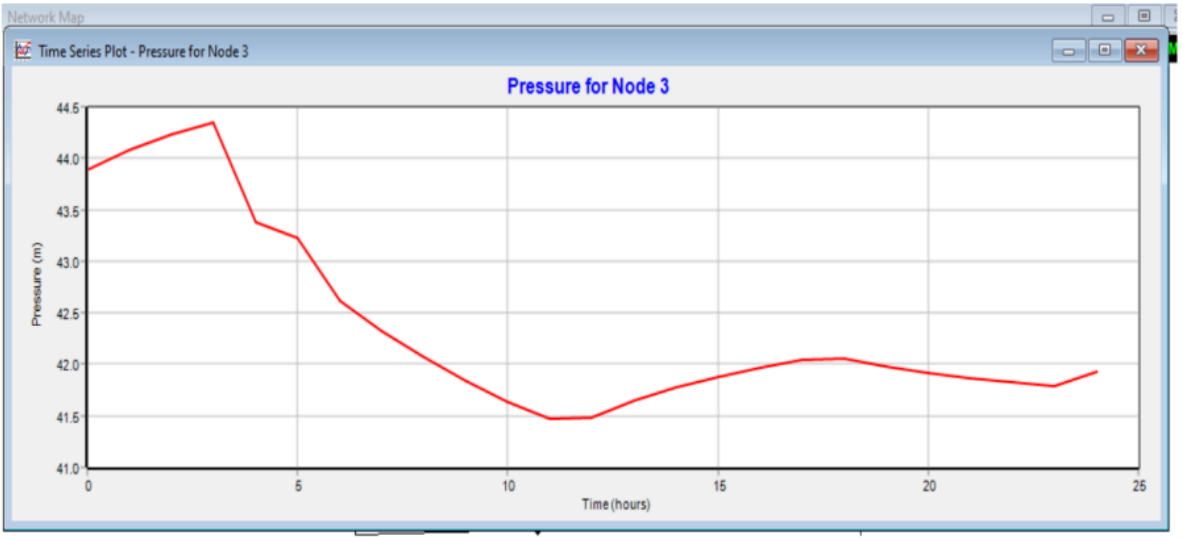


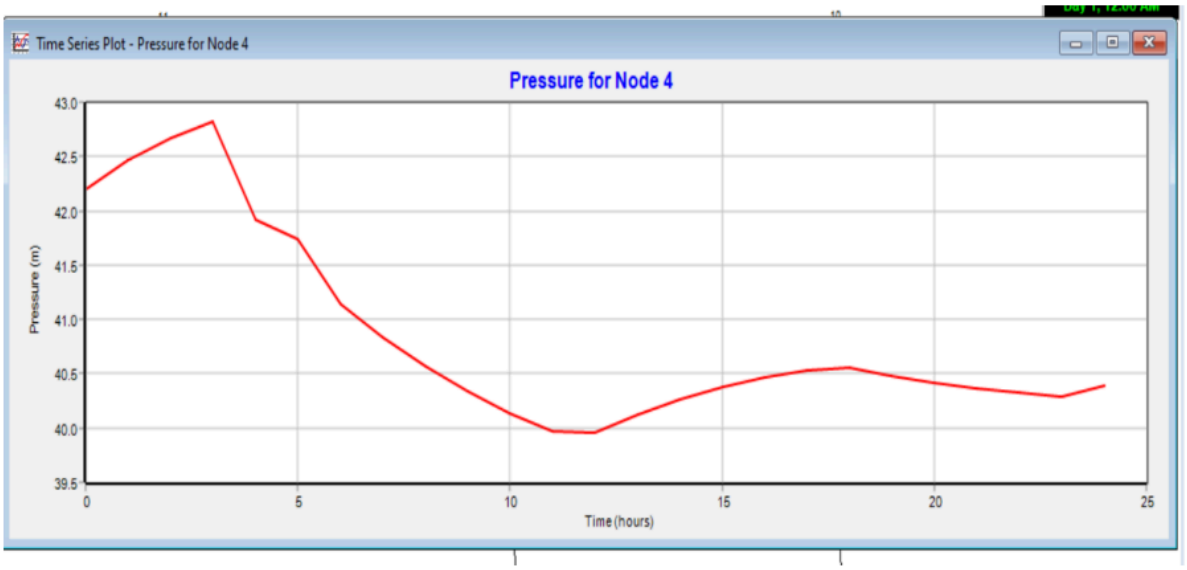


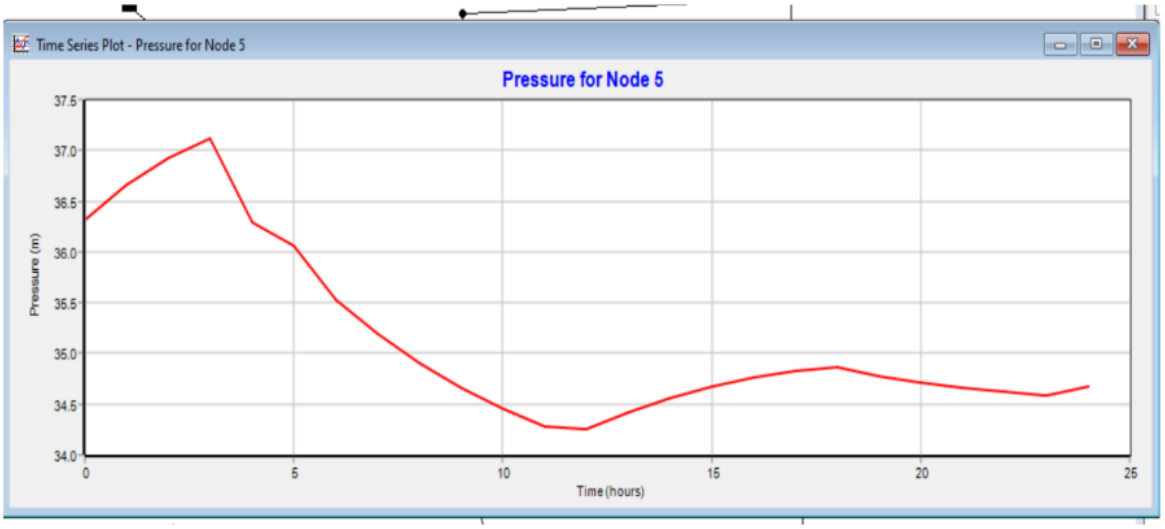
(b) time-series plots of pressure (in terms of water depth) at all the junctions.

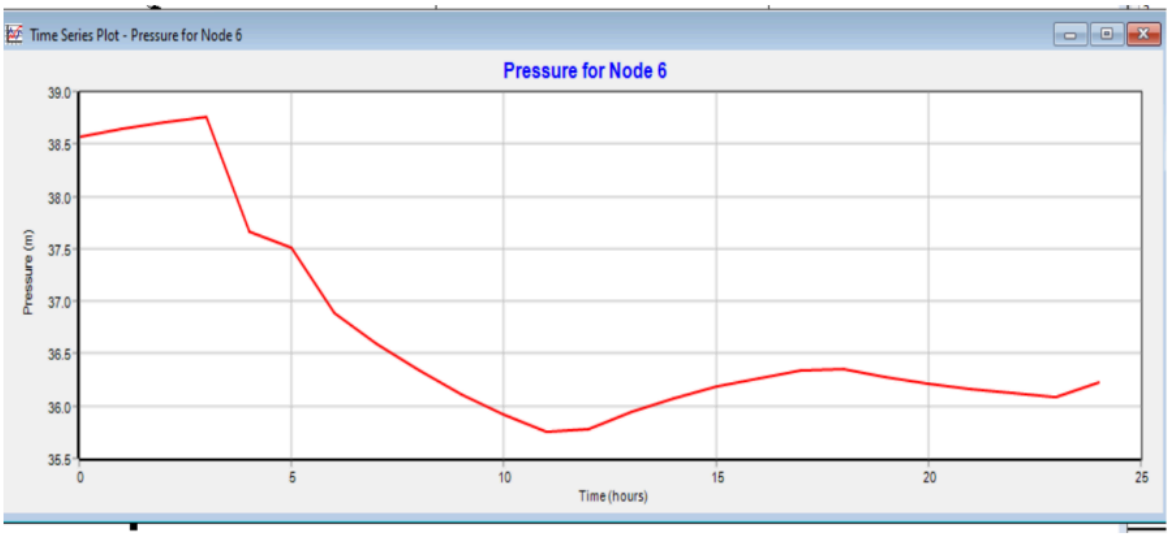


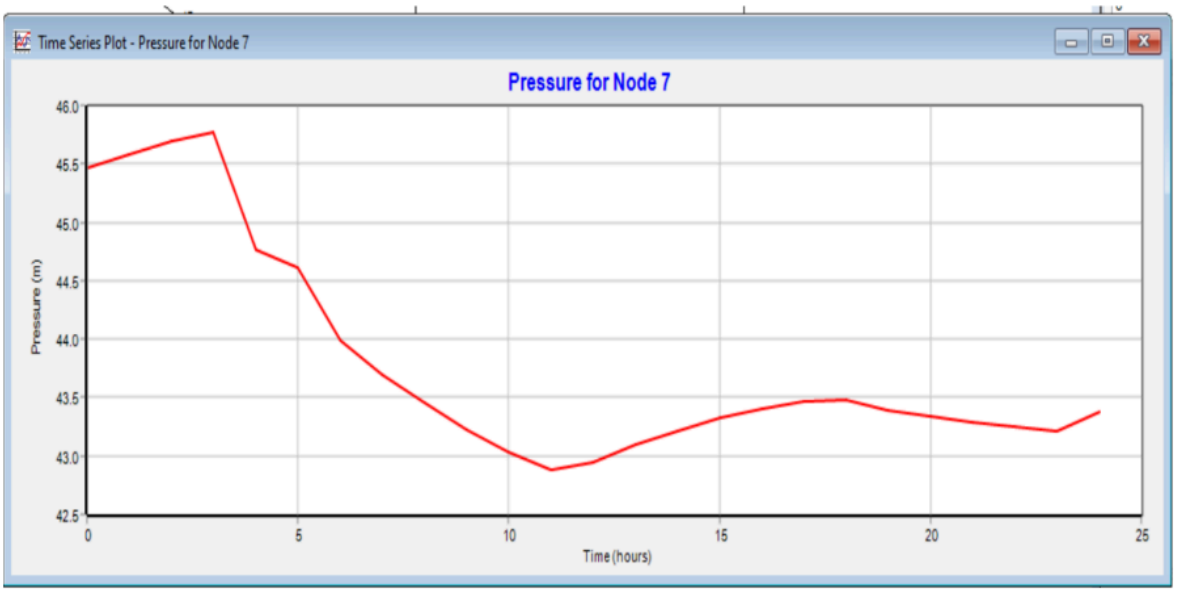

Time Series Plot - Pressure for Node 2
Pressure for Node 2
Pressure (m)
46.0
45.5
45.0
44.5
44.0
43.5
43.0
42.5
0
5
10
15
20
25
Time (hours)

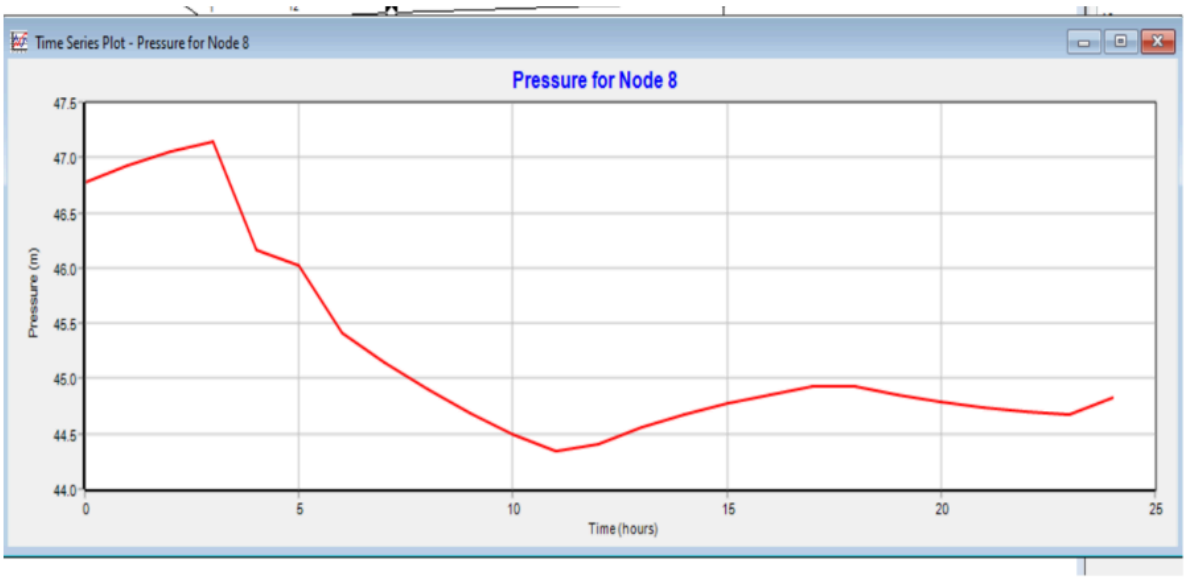

Network Map
Time Series Plot - Pressure for Node 3
Pressure for Node 3
Pressure (m)
44.5
44.0
43.5
43.0
42.5
42.0
41.5
41.0
0
5
10
15
20
25
Time (hours)

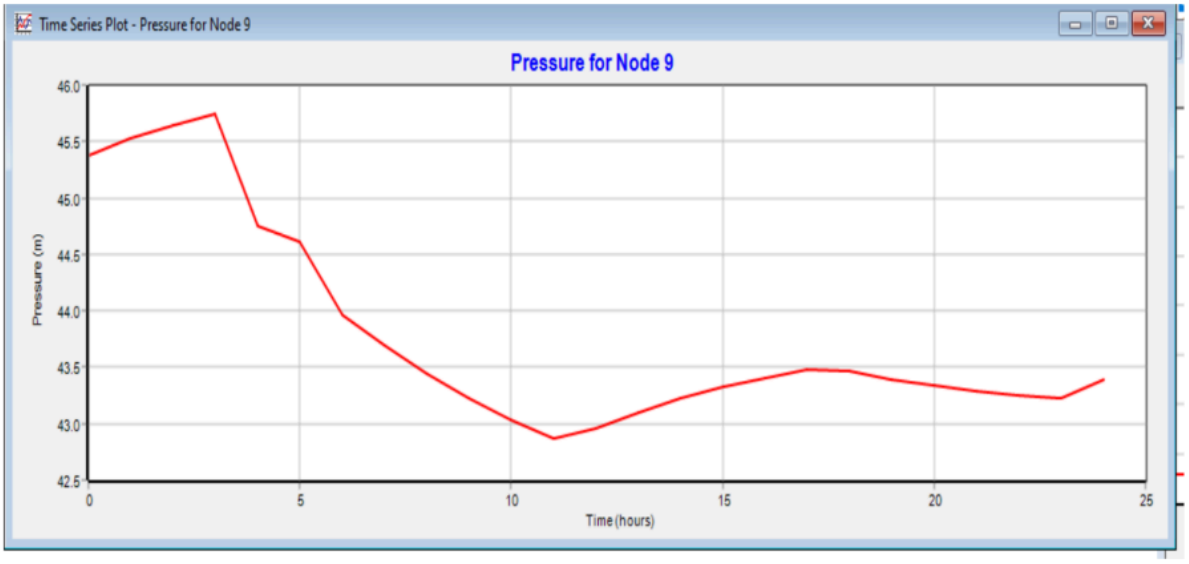

Time Series Plot - Pressure for Node 4
Pressure for Node 4
Pressure (m)
43.0
42.5
42.0
41.5
41.0
40.5
40.0
39.5
0
5
10
15
20
25
Time (hours)

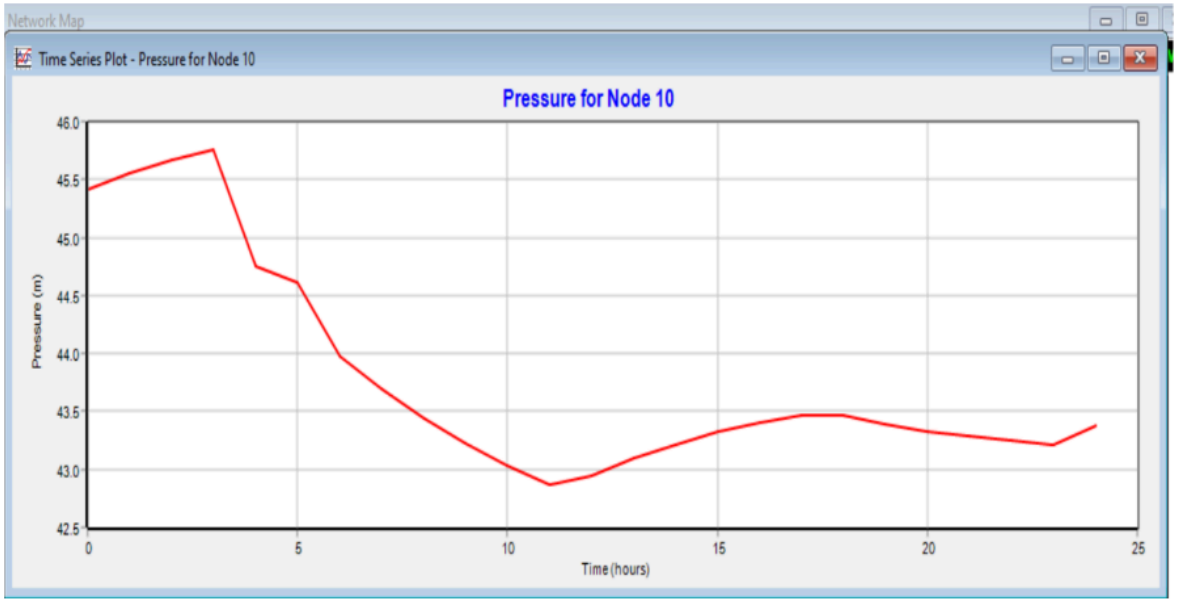

Time Series Plot - Pressure for Node 5
Pressure for Node 5
Pressure (m)
37.5
37.0
36.5
36.0
35.5
35.0
34.5
34.0
0
5
10
15
20
25
Time (hours)

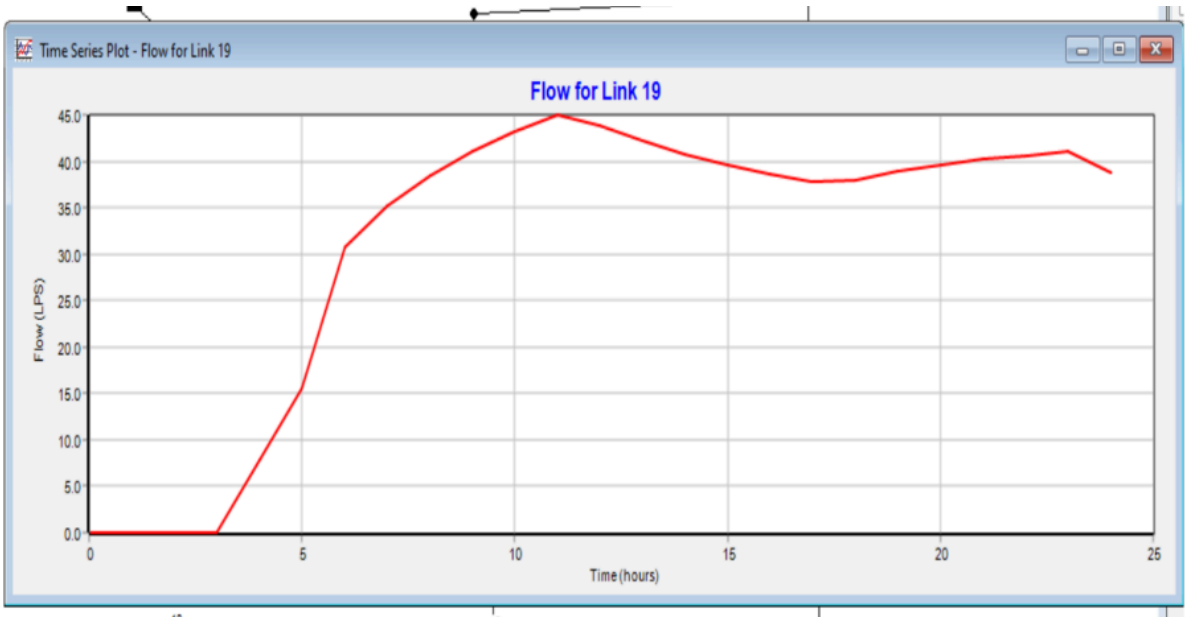

Time Series Plot - Pressure for Node 6
Pressure for Node 6
Pressure (m)
39.0
38.5
38.0
37.5
37.0
36.5
36.0
35.5
0
5
10
15
20
25
Time (hours)


Time Series Plot - Pressure for Node 7
Pressure for Node 7
Pressure (m)
46.0
45.5
45.0
44.5
44.0
43.5
43.0
42.5
0
5
10
15
20
25
Time (hours)


Time Series Plot - Pressure for Node 8
Pressure for Node 8
Pressure (m)
47.5
47.0
46.5
46.0
45.5
45.0
44.5
44.0
0
5
10
15
20
25
Time (hours)


Time Series Plot - Pressure for Node 9
Pressure for Node 9
Pressure (m)
46.0
45.5
45.0
44.5
44.0
43.5
43.0
42.5
0
5
10
15
20
25
Time (hours)


Network Map
Time Series Plot - Pressure for Node 10
Pressure for Node 10
Pressure (m)
46.0
45.5
45.0
44.5
44.0
43.5
43.0
42.5
0
5
10
15
20
25
Time (hours)


Time Series Plot - Flow for Link 19
Flow for Link 19
Flow (LPS)
45.0
40.0
35.0
30.0
25.0
20.0
15.0
10.0
5.0
0.0
0
5
10
15
20
25
Time (hours)